

Advanced Statistical Methods II

Tutorial I: Simulating Missingness in R

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Outline of Today's Tutorial

Tutorial orientation

Generating a complete dataset

MCAR

MAR

MNAR

Putting the mechanisms side by side

Practical workflow and key messages

Tutorial orientation

Companion to Lectures I and II

What this tutorial is for



Today we will create complete data, deliberately make values missing, and see what goes wrong when the missingness mechanism is ignored.

- $$R_i = \begin{cases} 1, & \text{if } Y_i \text{ is observed,} \\ 0, & \text{if } Y_i \text{ is missing.} \end{cases}$$

- A set of small navigation icons typically found in Beamer presentations, including symbols for back, forward, search, and other slide controls.

Generating a complete dataset

Start from a world where nothing is missing

The complete-data model for today



We simulate a simple linear regression setting:

$$X_i \sim N(0, 1), \quad Y_i = 2 + 1.5X_i + \varepsilon_i, \quad \varepsilon_i \sim N(0, 1).$$

- ▶ X_i : fully observed covariate.
- ▶ Y_i : outcome that may become missing.
- ▶ Complete-data regression coefficients are approximately 2 and 1.5.

We first create the truth. Then we hide some values of Y in different ways.

R setup: complete data



```
set.seed(2026)

n <- 1000
x <- rnorm(n, mean = 0, sd = 1)
eps <- rnorm(n, mean = 0, sd = 1)
y <- 2 + 1.5 * x + eps

dat0 <- data.frame(id = 1:n, x = x, y = y)

summary(dat0)
coef(lm(y ~ x, data = dat0))
```

The estimates of intercept is 1.984 and coefficient of X is 1.483.

A helper function for comparison



```
summarize_analysis <- function(data, yvar = "y_obs") {  
  ok <- !is.na(data[[yvar]])  
  fit <- lm(data[[yvar]][ok] ~ data$x[ok])  
  
  c(n_used      = sum(ok),  
    missing_pct = mean(!ok) * 100,  
    mean_y      = mean(data[[yvar]], na.rm = TRUE),  
    intercept   = coef(fit)[1],  
    slope       = coef(fit)[2])  
}
```

For each mechanism, we will ask: how many observations were lost, and how did the summary change?

A basic missingness plot in R



```
plot_missing <- function(data, yvar = "y_obs", main = "Missingness") {  
  miss <- is.na(data[[yvar]])  
  plot(data$x, data$y,  
        col = ifelse(miss, "red", "steelblue"),  
        pch = ifelse(miss, 4, 19),  
        xlab = "x", ylab = "true y",  
        main = main)  
  legend("topleft", c("observed y", "missing y"),  
        col = c("steelblue", "red"), pch = c(19, 4), bty = "n")  
}
```

This is a diagnostic plot for missing-data simulations. If you artificially create a y_{obs} column from y (e.g., by randomly or systematically setting some values to NA to simulate MCAR/MAR/MNAR mechanisms), this function lets you see the true y values that were suppressed, colored by whether they were "observed" or "missing."

In a real dataset, the true missing Y -values are not visible. In simulation, we can reveal them to learn what the mechanism is doing.

MCAR

Missing completely at random

MCAR means the missingness process does not depend on observed or unobserved data values.

$$\mathbb{P}(R_i = 0 \mid X_i, Y_i) = p.$$

- ▶ Every observation has the same probability of becoming missing.
- ▶ Missingness is like random deletion.
- ▶ Complete-case analysis loses precision, but it is not automatically biased.

R code: MCAR



```
dat_mcar <- dat0

p_miss <- 0.30
miss_mcar <- rbinom(n, size = 1, prob = p_miss) == 1

dat_mcar$y_obs <- dat_mcar$y
dat_mcar$y_obs[miss_mcar] <- NA

summarize_analysis(dat_mcar, "y_obs")
```

```
dat_mcar <- dat0

p_miss <- 0.30
miss_mcar <- rbinom(n, size = 1, prob = p_miss) == 1

dat_mcar$y_obs <- dat_mcar$y
dat_mcar$y_obs[miss_mcar] <- NA

summarize_analysis(dat_mcar, "y_obs")
```

Interpretation

About 30% of Y -values are removed, independently of X and Y .

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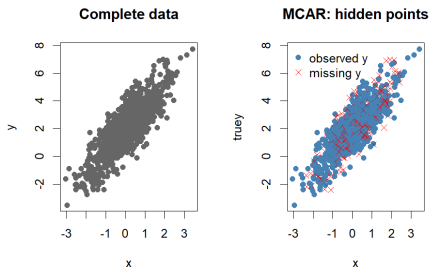
$\bar{y} = 1.940$, intercept term is 1.967 and slope is 1.463.

(originally: intercept was 1.984 and coefficient of X was 1.483.)

Visualizing MCAR



```
par(mfrow = c(1, 2))
plot(dat0$x, dat0$y, pch = 19, col = "gray40",
     xlab = "x", ylab = "y", main = "Complete data")
plot_missing(dat_mcar, "y_obs", "MCAR: hidden points")
par(mfrow = c(1, 1))
```



The missing values should be scattered without a systematic trend in X or Y .

MCAR: what usually happens?



- ▶ The observed cases still resemble a random subsample of the full dataset.
- ▶ Sample size is smaller, so standard errors usually increase.
- ▶ The sample mean and regression coefficients may fluctuate more.
- ▶ But systematic bias is not expected merely because values are MCAR.

MAR

Missing at random

MAR means the probability of missingness may depend on observed variables, but not on the missing value itself after conditioning on observed information.

$$\mathbb{P}(R_i = 0 \mid X_i, Y_i) = \mathbb{P}(R_i = 0 \mid X_i).$$

- ▶ Here X is observed for everyone.
- ▶ Units with larger X may be more likely to have missing Y .
- ▶ The marginal distribution of observed Y can be distorted.

R code: MAR depending on observed X



```
dat_mar <- dat0

# Probability of missingness depends on x, which is observed
prob_mar <- plogis(-1.2 + 1.3 * dat_mar$x)
miss_mar <- rbinom(n, size = 1, prob = prob_mar) == 1

dat_mar$y_obs <- dat_mar$y
dat_mar$y_obs[miss_mar] <- NA

summarize_analysis(dat_mar, "y_obs")
```

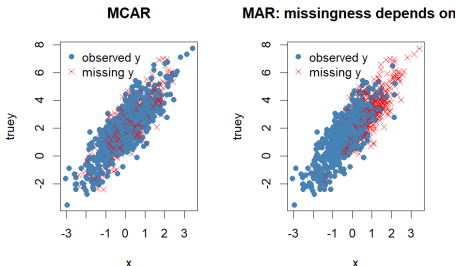
$\bar{y} = 1.519$, intercept term is 2.008 and slope is 1.517.

(originally: intercept was 1.984 and coefficient of X was 1.483.)

Why logistic probability?

$\text{plogis}(a + bX)$ keeps probabilities between 0 and 1 while letting missingness increase with X .

```
par(mfrow = c(1, 2))
plot_missing(dat_mcar, "y_obs", "MCAR")
plot_missing(dat_mar, "y_obs", "MAR: missingness depends on x")
par(mfrow = c(1, 1))
```



Unlike MCAR, the missing values now have a pattern connected to an observed variable.

MAR: implications for analysis



- ▶ The complete cases may overrepresent some parts of the X -space.
- ▶ The observed mean of Y can be biased for the population mean of Y .
- ▶ If the regression model $Y | X$ is correctly specified and X is observed, complete-case regression coefficients may still behave reasonably.

MNAR

Missing not at random

MNAR means missingness depends on the unobserved value itself, even after conditioning on observed information.

$\mathbb{P}(R_i = 0 \mid X_i, Y_i)$ depends on Y_i .

- ▶ Example: people with very poor health outcomes are less likely to respond.
- ▶ Example: high-income respondents are less willing to report income.
- ▶ The missing values are systematically different in a way the analyst cannot fully see.

R code: MNAR depending on true Y



```
dat_mnar <- dat0

# Probability of missingness depends on y itself.
# In real data, the missing y values would not be visible.
prob_mnar <- plogis(-2.0 + 0.8 * dat_mnar$y)
miss_mnar <- rbinom(n, size = 1, prob = prob_mnar) == 1

dat_mnar$y_obs <- dat_mnar$y
dat_mnar$y_obs[miss_mnar] <- NA

summarize_analysis(dat_mnar, "y_obs")
```

$\bar{y} = 1.241$, intercept term is 1.723 and slope is 1.367.

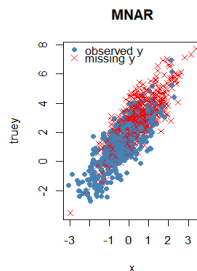
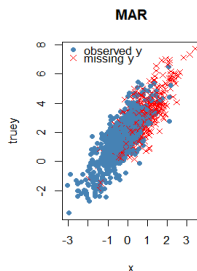
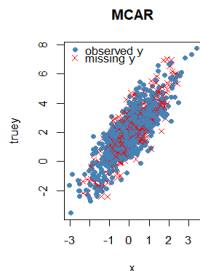
(originally: intercept was 1.984 and coefficient of X was 1.483.)

This code uses the true Y only because this is a simulation. In real data, this mechanism is exactly what makes the problem difficult.

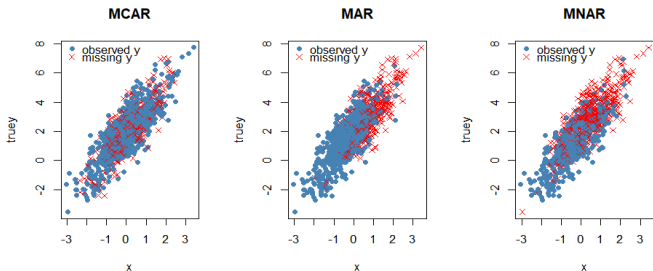
Visualizing MNAR



```
par(mfrow = c(1, 3))
plot_missing(dat_mcar, "y_obs", "MCAR")
plot_missing(dat_mar, "y_obs", "MAR")
plot_missing(dat_mnar, "y_obs", "MNAR")
par(mfrow = c(1, 1))
```



```
par(mfrow = c(1, 3))
plot_missing(dat_mcar, "y_obs", "MCAR")
plot_missing(dat_mar, "y_obs", "MAR")
plot_missing(dat_mnar, "y_obs", "MNAR")
par(mfrow = c(1, 1))
```



Here high values of Y are more likely to disappear. The observed data may make the outcome look artificially low.

MNAR: why this is especially hard



- ▶ The mechanism involves values that are not observed.
- ▶ Purely observed-data plots may not reveal the full problem.
- ▶ Standard methods based on MCAR or MAR may give misleading answers.
- ▶ Sensitivity analysis and subject-matter knowledge become important.

Can we prove from the observed data alone that the mechanism is MNAR? Usually, no.

Putting mechanisms side by side

Same percentage missing, different statistical meaning

R code: compare all mechanisms



```
res <- rbind(  
  complete = summarize_analysis(transform(dat0, y_obs = y), "y_obs"),  
  MCAR      = summarize_analysis(dat_mcar, "y_obs"),  
  MAR       = summarize_analysis(dat_mar, "y_obs"),  
  MNAR      = summarize_analysis(dat_mnar, "y_obs")  
)  
  
round(res, 3)
```

```
> round(res, 3)
```

	n_used	missing_pct	mean_y	intercept.(Intercept)	slope.data\$x[ok]
complete	1000	0.0	2.004	1.984	1.483
MCAR	699	30.1	1.940	1.967	1.463
MAR	703	29.7	1.519	2.008	1.517
MNAR	557	44.3	1.241	1.723	1.367

Conceptual comparison



Mechanism	Missingness depends on	de- Typical implication
MCAR	nothing in the data	loss of sample size and precision
MAR	observed variables	bias possible for naive marginal summaries; model-based methods may help
MNAR	unobserved/missing values	potentially serious bias; sensitivity analysis often needed

The mechanism matters as much as, and sometimes more than, the percentage missing.

$$\text{logit}\{\mathbb{P}(R_i = 0)\} = \alpha_0 + \alpha_1 X_i.$$

A simple diagnostic regression is:

$$\text{logit}\{\mathbb{P}(R_i = 0)\} = \alpha_0 + \alpha_1 X_i.$$

```
dat_mar$miss <- as.integer(is.na(dat_mar$y_obs))  
fit_miss <- glm(miss ~ x, data = dat_mar, family = binomial)  
summary(fit_miss)
```


A simple diagnostic regression is:

$$\text{logit}\{\mathbb{P}(R_i = 0)\} = \alpha_0 + \alpha_1 X_i.$$

```
dat_mar$miss <- as.integer(is.na(dat_mar$y_obs))  
fit_miss <- glm(miss ~ x, data = dat_mar, family = binomial)  
summary(fit_miss)
```

This checks association between missingness and observed variables. It does not determine the full missing-data mechanism.

Practical workflow and key messages

A practical first workflow in R



1. Identify which variables have missing values.
2. Create missingness indicators.
3. Compare observed covariates between missing and non-missing groups.
4. Decide what estimand is affected: mean, regression coefficient, prediction, or likelihood.
5. Avoid declaring MCAR/MAR/MNAR purely from software output.
6. Record assumptions clearly before choosing a method.

Good missing-data analysis begins before imputation or modeling: it begins with the missingness process.

- ▶ Change p_{miss} from 0.30 to 0.10, 0.50, and 0.70.
- ▶ Change the MAR model from $-1.2 + 1.3 \cdot x$ to $-1.2 - 1.3 \cdot x$.
- ▶ Change the MNAR model so that low values of Y are more likely to be missing.
- ▶ Run each simulation 100 times and summarize the average bias.

Which matters more in your experiments: the missingness rate or the mechanism?

Appendix: compact R script, part 1



```
set.seed(2026)

n <- 1000
x <- rnorm(n)
eps <- rnorm(n)
y <- 2 + 1.5 * x + eps

dat0 <- data.frame(id = 1:n, x = x, y = y)

summarize_analysis <- function(data, yvar = "y_obs") {
  ok <- !is.na(data[[yvar]])
  fit <- lm(data[[yvar]][ok] ~ data$x[ok])
  c(n_used      = sum(ok),
    missing_pct = mean(!ok) * 100,
    mean_y      = mean(data[[yvar]], na.rm = TRUE),
    intercept   = coef(fit)[1],
    slope       = coef(fit)[2])
}
```

Appendix: compact R script, part 2



```
# MCAR
dat_mcar <- dat0
miss_mcar <- rbinom(n, 1, 0.30) == 1
dat_mcar$y_obs <- dat_mcar$y
dat_mcar$y_obs[miss_mcar] <- NA

# MAR
dat_mar <- dat0
prob_mar <- plogis(-1.2 + 1.3 * dat_mar$x)
miss_mar <- rbinom(n, 1, prob_mar) == 1
dat_mar$y_obs <- dat_mar$y
dat_mar$y_obs[miss_mar] <- NA

# MNAR
dat_mnar <- dat0
prob_mnar <- plogis(-2.0 + 0.8 * dat_mnar$y)
miss_mnar <- rbinom(n, 1, prob_mnar) == 1
dat_mnar$y_obs <- dat_mnar$y
dat_mnar$y_obs[miss_mnar] <- NA
```

Appendix: compact R script, part 3



```
res <- rbind(
  complete = summarize_analysis(transform(dat0, y_obs = y), "y_obs"),
  MCAR      = summarize_analysis(dat_mcar, "y_obs"),
  MAR       = summarize_analysis(dat_mar, "y_obs"),
  MNAR      = summarize_analysis(dat_mnar, "y_obs")
)
round(res, 3)

plot_missing <- function(data, yvar = "y_obs", main = "Missingness") {
  miss <- is.na(data[[yvar]])
  plot(data$x, data$y,
    col = ifelse(miss, "red", "steelblue"),
    pch = ifelse(miss, 4, 19),
    xlab = "x", ylab = "true y", main = main)
  legend("topleft", c("observed y", "missing y"),
    col = c("steelblue", "red"), pch = c(19, 4), bty = "n")
}
```